6 METALS

Look at these pictures... They provide a glimpse of the modern world. Trains running on electricity, the prominent changes in the field of construction, the diversity of electronic technology etc. have all changed our lives drastically. You might have seen in news clips how the army quickly build the Bailey bridge on the site of a disaster. Can we think of a world without cars, smartphones, computers, internet, electricity, buildings or bridges? For this, we are indebted to a special class of elements. Do you know which class it is?

The discovery of metals and their role in human progress is amazing. The term 'metal age' refers to the transition period from the stone age to the use of metals. Our ancient period is marked by the names copper age, bronze age, iron age etc. Metals are used in many forms in different fields of our life.

Try to find some of them and complete the table.

Transport sector	Cars, cycles, ..., ...
Agricultural sector	Agricultural tools, tillers, ..., ...
Construction sector	Buildings, bridges, pipes ....., ...
Technology sector	Phones, laptops, electrical wires ....., ...
Consumer goods	Jewellery, household articles ....., ...

Table 6.1

Which metals are mostly used in these situations? List them.

Iron, gold, ..., ..., ...,

You have learnt that metals are used in various fields because of some of their common properties.

What properties of metals are utilised in various fields?

- Sonority
- Malleability
- Ductility
-
-

Bailey bridge

Bailey bridge is an emergency bridge constructed in disaster stricken areas temporarily. Such bridges are also used for military purposes. It is a kind of portable bridge that can be easily constructed by assembling prefabricated parts. Steel and wood are the main components of the bridge. The role played by such metal bridges in rescue operations in disaster stricken regions is very important. Such a bridge was first built by a civil engineer Donald Coleman Bailey, an officer in the British War Office, during World War II and the bridge was named after him.

Occurrence of metals in nature

The earth's crust is the major source of metals. Metals are generally found in combined state as they are highly reactive. Metal compounds like sodium chloride and magnesium chloride are dissolved in sea water also. A very few metals are seen in free state too. Can you identify these? .....

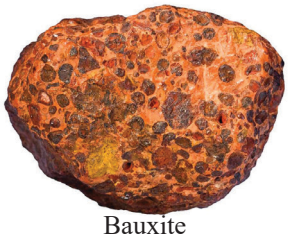
Naturally occurring metals or compounds are generally called minerals. These include metallic compounds and non metallic compounds.

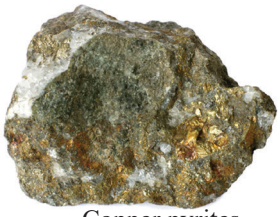
The same metal can be found in the form of various minerals in nature. Metals are separated from the most suitable ones among these. For example, clay ($Al_2O_3.2SiO_2.2H_2O$) is an abundant mineral of aluminium. Bauxite ($Al_2O_3.2H_2O$) and cryolite (Na_3AlF_6) are also minerals of aluminium. But among these, bauxite is the mineral that is commonly used to produce aluminium. What will be the reason?

What are the qualities of a mineral which is used to extract a metal?

- Abundance
- Ease of extraction
- High metal content
- Low production cost.

Based on these characteristics, bauxite is the most suitable mineral for the production of aluminium.


Bauxite


Copper pyrites

Haematite

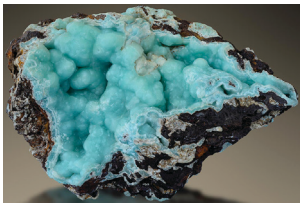

Calamine

Fig. 6.1

Any mineral from which a metal can be extracted easily and economically is called an ore of that metal.

The ores of some metals are listed in Table 6.2.

Metal	Ore	Chemical formula
Aluminium(Al)	Bauxite	$Al_2O_3.2H_2O$
Iron (Fe)	Haematite	Fe_2O_3
	Magnetite	Fe_3O_4
Copper (Cu)	Copper pyrites	$CuFeS_2$
	Copper glance	Cu_2S
	Cuprite	Cu_2O
Zinc (Zn)	Zinc blende	ZnS
	Calamine	$ZnCO_3$
Tin (Sn)	Cassiterite / Tinstone	SnO_2

Table 6.2

Highly reactive metals are found in nature in their combined form. Let us examine how pure metals are extracted from these.

The process of extraction of metal from ores involves three major stages. Such processes are generally referred to as metallurgy.

Concentration of ores	Extraction of metal from concentrated ore	Refining of metal

Concentration of ores

Ores are mined from the earth's crust. A lot of earthy impurities like soil and sand may be mixed with this. Concentration of ores is the process of increasing the metal content of the ore by removing such impurities, known as gangue. Depending on the nature of the ore and gangue, various methods are adopted for this purpose.

1. Levigation or Hydraulic washing

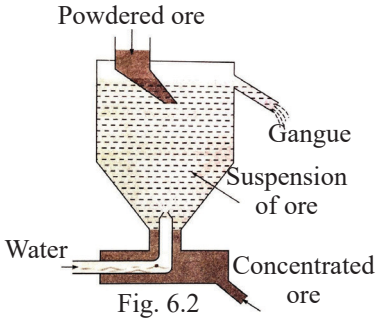


Fig. 6.2

This method is employed when the density of the ore is more than that of the gangue. The powdered ore is washed in a stream of water. The less dense gangue, which floats, is filtered out. The more dense ore remains at the bottom. Generally, oxide ores can be concentrated by this method.

2. Froth floatation

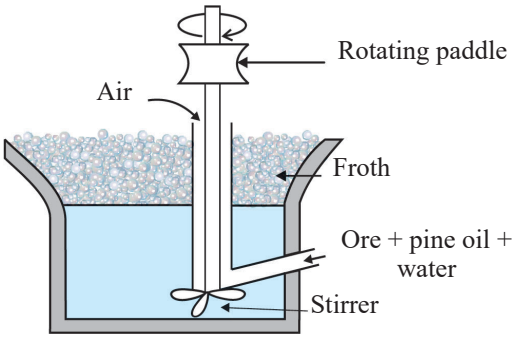


Fig. 6.3

The froth floatation process is used when the ore is less dense than the gangue. The powdered ore is added to a mixture of water and pine oil and is stirred in a strong current of air. The ore particles stick to the froth formed, and float. It is removed and then the ore is separated and dried.

Generally, sulphide ores are concentrated by this method.

3. Magnetic separation

When the powdered ore is passed through a magnetic roller, either the ore or the impurities having magnetic property is separated. In this way, pure ore is obtained by magnetic separation.

For example, magnetite (Fe_3O_4) ore has magnetic properties, but the gangue is non magnetic. Tinstone (SnO_2) ore is non magnetic, but iron tungstate, the gangue present in it is magnetic.

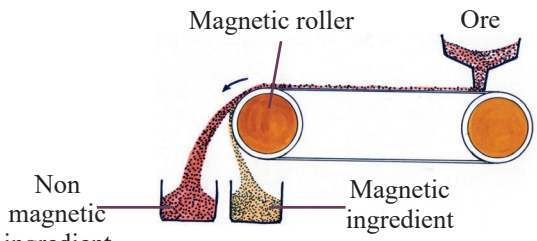


Fig. 6.4

4. Leaching

The powdered ore is mixed with a suitable solvent. The ore dissolves in it or undergoes a chemical reaction to become a solution. The gangue remains insoluble in it. It is then filtered out. This process is called leaching.

Bauxite, the ore of aluminium, is concentrated in this way.

Metals such as gold and silver occur almost freely in nature. The ores of these metals are leached with a dilute solution of sodium cyanide (NaCN) or potassium cyanide (KCN) in the presence of air to extract the metals.

The nature of ores is given. Complete the table using appropriate concentration methods.

Nature of ore	Concentration method
1. Ores are less dense than the impurities.	.....
2. Ores have magnetic properties but impurities are non magnetic.	.....
3. A solvent which dissolves the ore is used.	.....
4. Ores are denser than the impurities.	.....

Extraction of metal from concentrated ore

You have learnt that the reactivity of metals differ. You also know about the reactivity series that is prepared based on this.

How are metals arranged in the reactivity series?

Decreasing order of reactivity / increasing order of reactivity

Is there any relation between the production of metal and the reactivity series?

Here are some metals in the reactivity series.

Reactivity decreases ↓

Potassium (K) Sodium (Na) Calcium (Ca) Magnesium (Mg) Aluminium (Al)	Metals that are highly reactive and form stable compounds.
Zinc (Zn) Iron (Fe) Tin (Sn) Lead (Pb)	Metals with comparatively moderate reactivity and form stable compounds.
Copper (Cu) Mercury (Hg) Silver (Ag) Gold (Au)	Less reactive metals and form less stable compounds. They occur more or less in the free state in nature.

Mercury can be easily separated from its ores by controlled heating.

Metals with comparatively moderate reactivity like Zn, Fe, Sn, Pb, are generally found as oxides, sulphides or carbonates.

Eg. $ZnCO_3$, ZnS, Fe_2O_3, SnO_2, PbS.

There are two stages for the production of metal from them.

1. Conversion of concentrated ore to oxide

Two methods are commonly used for this.

(a) Calcination

Metal carbonates or hydroxides are converted to their oxides by a process called calcination.

Calcination refers to heating the concentrated ore in limited quantities or absence of air at a temperature below the melting point.

Eg. $CaCO_3(s) \xrightarrow{heat} CaO(s) + CO_2(g)$

$ZnCO_3(s) \xrightarrow{heat}$ ..... + ...

(b) Roasting

Roasting refers to heating the concentrated ore at a temperature below the melting point in the presence of air.

Zinc sulphide and lead sulphide can be easily converted into their oxides by this method.

$2ZnS(s) + 3O_2(g) \xrightarrow{heat} 2ZnO(s) + 2SO_2(g)$

2PbS(s) + ... $\xrightarrow{heat}$ ..... + ...

When the concentrated ore is subjected to roasting, the water content in them gets released as vapours. They become pure and moisture free oxide ores.

2. Reduction of oxide ore

The metal part of the ores is in the form of positive ions, and hence it can be extracted by a reduction process.

Note the process that separates zinc from zinc oxide.

$$ZnO + C \rightarrow Zn + CO$$

Which is the reducing agent in this case?

.....

Reducing agents such as coke (an allotrope of carbon) and carbon monoxide are used to extract moderately reactive metals such as iron, zinc and tin from their oxide ores. Carbon has a greater tendency to react with oxygen at high temperatures, so oxygen is easily removed from the ore.

Strong reducing agents are required to extract highly reactive metals such as sodium, potassium, calcium, magnesium and aluminium from their ores. Such metals are produced by electrolysis.

Complete Table 6.3 regarding the manufacture of metals.

Method of separating metals from ores	Metals
Electrolysis	.....
Reduction using coke	.....
Heating in controlled ways	.....

Table 6.3

The order of reactivity of some metals is given.

Al > Zn > Fe > Au

- Which metal forms the most stable compound?
- Which metal is produced by electrolysis?
- Which metal can displace zinc from compounds of zinc? Why?
- Which metal is found in the free state in nature?
- Which metal is produced by reduction using carbon?

Refining of metals

The metal extracted from concentrated ore may contain other elements or their compounds as impurities. Refining of metals is the process of removal of these impurities to produce pure metal.

Various methods are adopted for this purpose depending upon the nature of the metal to be purified and the impurities contained in it. Some are given below.

(a) Liquation

When metals with low melting point contain impurities with high melting point, the metal is heated on the inclined surface of the furnace. The pure metal separates from the impurities, melts, flows down and it is collected. This is known as liquation. Metals such as tin (Sn) and lead (Pb) having low melting point can be purified by this method.

(b) Distillation

This method is used to purify metals with relatively low boiling points. When the metal containing impurities is heated under suitable conditions, only the pure metal vapourises. When this vapour condenses, pure metal is obtained.

Metals like zinc (Zn), cadmium (Cd) and mercury (Hg) can be purified using this method.

(c) Electrolytic refining

In an electrolytic cell, an impure metal is used as the anode and a thin piece of pure metal is used as the cathode. A suitable salt solution of the same metal is used as the electrolyte. When electric current is passed, the pure metal alone is separated from the anode and deposited at the cathode.

Anode : $M \rightarrow M^{n+} + ne^-$

Cathode : $M^{n+} + ne^- \rightarrow$ .....

(Hint : M - metal, n - number of electrons involved in the reaction)

Copper (Cu) and zinc (Zn) can be purified in this way.

Now, let us get familiar with the manufacturing methods of some industrially important metals.

Extraction of iron

Iron comprises about 5% of the earth's crust and it is the second most abundant metal.

Complete Table 6.4 regarding the manufacture of iron.

Ores of iron and its chemical formula	.....
Methods used for concentration	Washing in stream of water, roasting, ...
Substance used to reduce the ore	.....

Table 6.4

Iron is produced in steel towers, known as blast furnaces, which are 25 to 30 meters high. Its interior is lined with bricks (Refractory bricks) capable of withstanding high temperatures (Figure 6.5). The concentrated ore mixed with limestone ($CaCO_3$) and coke (C) is fed from the top of the furnace via cup & cone arrangement. This mixture is called charge.

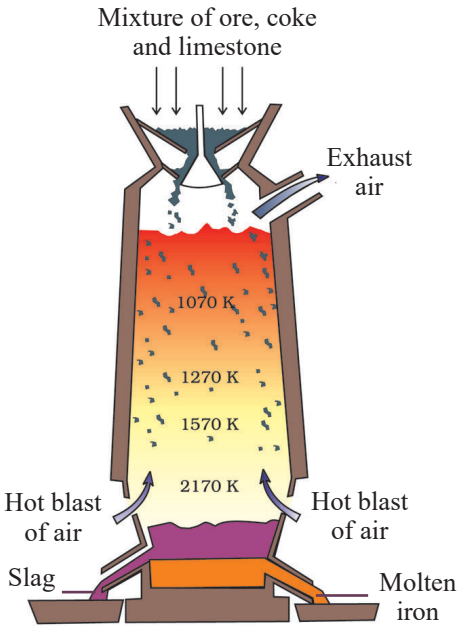


Fig. 6.5

A blast of hot air is blown into the furnace from its bottom. As the charge descends from the top of the furnace, various chemical changes take place inside it. The coke combines with the oxygen in the air, when the blast of hot air flows from the bottom to the top of the blast furnace.

$$C + O_2 \rightarrow CO_2 + \text{heat}$$

This is an exothermic reaction.

Coke reacts with carbon dioxide to produce carbon monoxide.

This is an endothermic reaction.

$$CO_2 + C + \text{heat} \rightarrow 2CO$$

The main chemical reaction taking place in the blast furnace is the reduction of iron ore using carbon monoxide.

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

In the production of iron, limestone is added to the ore. What could be the reason?

Even though the iron ore fed into the furnace is in the most refined form, it may contain acidic impurities such as sand (silica SiO_2) that cannot be removed in the first stages of concentration methods. For this, a substance known as flux is added during the metallurgical stage. As the impurities are acidic in nature, a basic substance is used as the flux. The flux combines with gangue to form molten slag.

$$\text{Gangue} + \text{Flux} \rightarrow \text{Slag}$$

When heated at high temperature, the limestone decomposes to calcium oxide (Quick lime). Calcium oxide produced in this reaction acts as flux.

Let us examine the chemical changes of limestone inside the blast furnace.

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

Calcium oxide reacts with sand (SiO_2) to form calcium silicate (slag) which melts easily.

$$\underset{\text{Flux}}{CaO(s)} + \underset{\text{Gangue}}{SiO_2(s)} \rightarrow \underset{\text{Slag}}{CaSiO_3(l)}$$

If the gangue in the ore is basic in nature, what type of flux should be used? .....

Since the molten slag is less dense, it floats on the molten iron. These are removed at regular intervals through a separate tube / tap. This slag is used in the production of cement and in the construction of roads.

The molten iron obtained from the blast furnace contains 4% carbon and small amounts of other impurities such as manganese, silicon, phosphorus and sulphur. This is called pig iron.

If the percentage of impurity in it is reduced, the iron obtained can be cast as desired. It is called cast iron. It contains more than 2% carbon.

Wrought iron is a relatively pure form of iron, containing 0.02% - 0.05% carbon and small amounts of phosphorus, silicon, manganese, sulphur etc.

Steel

The pig iron obtained from the blast furnace is not suitable either for hammering into various shapes or for drawing into wires. The steel obtained from this is used for various industrial purposes. Steel is iron containing 0.05% to 1.5% carbon. There are different types of carbon steels depending on the amount of carbon. They differ in properties also (Table 6.5).

Carbon steel	Amount of carbon (%)	Features
Mild steel	0.05 - 0.2	Easy to draw into wires and hammer into plates. Strong and hard. Used for making agricultural tools.
Medium steel	0.2 - 0.6	High hardness. Used for the construction of railway tracks, handrails, rafters etc.
High carbon steel	0.6 - 1.5	Very high elasticity and hardness. Used for the manufacture of surgical instruments, springs, knives, drills etc.

Table 6.5

Prepare a list of different types of steels, their methods of manufacture and use. ICT can be utilised.

Alloy steel

The use of metals in their pure form is relatively less. Most of the metals are converted into alloys to be used for various purposes in daily life. List the alloys familiar to you.

Brass, ……., ……, ……..

Alloys are mixtures of two or more metals. Nonmetals like carbon, nitrogen and phosphorus are also used for the production of alloys.

What are the constituents of steel? Iron, ……..

Alloy steels are made by adding other metals to steel. Their properties are different from those of steel.

Let us get familiar with some important alloy steels.

Alloy steels	Constituent elements	Properties and uses
Stainless steel	Fe, Cr, Ni, C	Resists corrosion. Manufacture of household appliances and blades.
Manganese steel	Fe, Mn, C	High hardness. Used for the manufacture of drills, safes, plates and railway tracks.
Silicon steel	Fe, Si, C	Also known as electrical steel. As it reduces electric loss, it is used in the core of electromagnetic instruments like motors, generators, transformers etc.

Table 6.6

Expand the list by finding more alloys, their constituents and uses.

Extraction of zinc

Zinc is another industrially important metal. Complete Table 6.7 regarding the production of zinc.

Sulphide ore of zinc (Common name, chemical formula)	.....
Concentration method	.....
Carbonate ore of zinc (Common name, chemical formula)	.....

Table 6.7

What are the processes used in the conversion of zinc ore to zinc oxide after the concentration of ore?

.....

Coke is mixed with the zinc oxide obtained after these processes.

This mixture is put into a furnace similar to a blast furnace. At high temperature in the furnace, the coke reduces zinc oxide to zinc.

$$ZnO(s) + C(s) \xrightarrow{heat} Zn(s) + CO(g)$$

Zinc has comparatively low boiling point (907°C) and hence it vapourises at high temperature and comes out. It is immediately cooled to liquid form.

You are familiar with the purification process of zinc. What is it known as?

.....

Brass is an important alloy containing zinc.

Prepare a note on the production of zinc and its use. Record it in the science diary.

Extraction of copper

Principal ores of copper

Copper pyrites - $CuFeS_2$ Copper glance - Cu_2S

What method of concentration is suitable for these sulphide ores?

.....

The ore contains basic impurities such as iron oxide even after concentration. What type of flux is suitable to remove it?

Acidic / Basic

Sand (SiO_2) having acidic nature is added to the ore and heated to form cuprous sulphide (Cu_2S) of maximum purity. It is partially converted to cuprous oxide (Cu_2O) when heated under controlled flow of air. The cuprous oxide converts the remaining cuprous sulphide into copper.

$$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2\uparrow$$

Note that no reducing agent is added here.

Molten copper is cooled in moulds. The copper lumps thus obtained are called blister copper. Why is it called so? The sulphur dioxide gas produced during the production of copper is released when the metal condenses. As a result blisters are formed and they appear as such on the surface of copper lumps.

Brass and bronze are important alloys that contain copper. List the constituent metals in them.

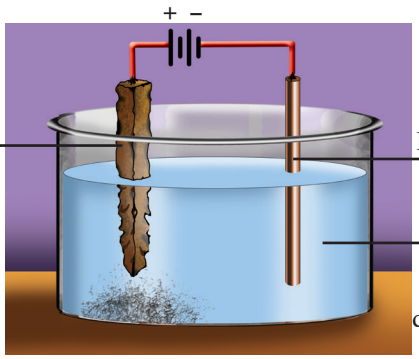


Fig. 6.6

Purification of Copper

You are familiar with the method of refining copper. Copper used for electrical purposes should be 99.99% pure. A high quality product is obtained through electrolytic refining. The diagram of refining of copper is given (Figure 6.6).

Observe the diagram and complete Table 6.8.

Anode	
Cathode	
Electrolyte	
Chemical equation of the reaction taking place at anode.	
Chemical equation of the reaction taking place at cathode.	

Table 6.8

Prepare a note on the process involved in the purification of copper and record it in the science diary.

Aluminium (Al)

Aluminium is the most abundant metal in the earth's crust. It is almost 8.3%. In 1827, the German scientist Friedrich Wohler extracted the metal from aluminium chloride using potassium. In those days, aluminium was more expensive than gold due to high production cost and difficulty in manufacturing. Aluminium was rarely used at that time. Later, when it was discovered that aluminium can be separated by electrolysis, it became an easily available metal. In 1886, the American scientist Charles Martin Hall and the French scientist Paul Heroult, through independent efforts, discovered a method to extract aluminium from bauxite. As the credit of discovering this method goes to both of them, the process is called Hall-Heroult process.

Charles Martin Hall
1863 - 1914

Paul Heroult
1863 - 1914

Extraction of aluminium

Aluminium is produced by Hall-Heroult process. It has two stages.

1. Concentration of bauxite

You have learnt that bauxite is concentrated by leaching. The powdered ore is treated with hot concentrated sodium hydroxide (NaOH) solution. The ore alone dissolves in it to form sodium aluminate ($NaAlO_2$) solution. Impurities are then filtered off.

$$Al_2O_3.2H_2O + 2NaOH \rightarrow 2NaAlO_2 + 3H_2O$$

A little aluminium hydroxide ($Al(OH)_3$) is added to the solution of sodium aluminate and stirred vigorously. It is diluted by adding water. As a result, pure aluminium hydroxide separates out from the solution.

$$NaAlO_2 + 2H_2O \rightarrow Al(OH)_3 + NaOH$$

If the aluminium hydroxide thus obtained is heated strongly, anhydrous alumina (Al_2O_3) is obtained.

$$2Al(OH)_3 \xrightarrow{heat} Al_2O_3 + 3H_2O$$

2. Electrolysis of aluminium oxide

Molten cryolite (Na_3AlF_6) is added to the alumina, obtained after concentration, to get a solution. This is the electrolyte used in the production of aluminium. The melting point of alumina is 2017°C. Cryolite is added to reduce the melting point and to increase the electrical conductivity.

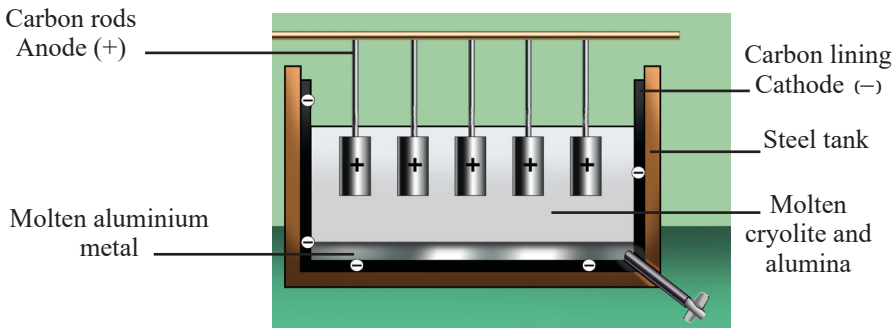


Fig. 6.7

Aluminium is separated at the cathode when a high voltage electric current is passed through the molten mixture. Examine the chemical equations of the reactions.

$$Al_2O_3 \rightarrow 2Al^{3+} + 3O^{2-}$$

What reaction takes place at the cathode?

$$Al^{3+} + 3e^- \rightarrow .....$$

What reaction takes place at the anode?

$$2O^{2-} \rightarrow O_2(g) + 4e^-$$

Which gas is liberated at the anode? .....

In the electrolytic cell, the positively charged carbon rods are replaced periodically. What is the reason?

Winged Metal (Duralumin)

At the beginning of the 20th century, a German scientist Alfred Wilm made an alloy of aluminium by adding copper, manganese and magnesium. Constant experimentation was undertaken to improve its strength and quality, and a new alloy was produced in Duren in 1911 by adding accurate amount of constituent metals and heating them at suitable temperature. It was named duraluminium by adding the name of the place. Later, the name was changed to duralumin. In 1919, duralumin began to be used in the construction of airplanes. Duralumin is used as an alloy in the manufacture of aircraft and aerospace vehicles and therefore be described as winged metal. The components of duralumin are Al-95.5%, Cu-3%, Mg-1% and Mn-0.5%. The duralumin produced in this way has the same strength as that of steel.

Oxygen released at the anode oxidises the carbon rod to oxides of carbon. As a result, the size of the anode rod decreases.

Complete Table 6.9 related to electrolysis of aluminium.

Anode	
Cathode	
Electrolyte	
The product formed at anode	
The product formed at cathode	
Reducing agent	

Table 6.9

Duralumin and alnico are important alloys which contain aluminium. The constituents of alnico are Fe, Al, Ni, Co etc. Alnico is used for making permanent magnets.

Corrosion of metals

Corrosion of metal is a process in which the metal reacts with a surrounding medium and undergoes chemical change. Corrosion of iron has been discussed in previous classes.

What are the factors influencing the corrosion of iron?

.....

The metals occupying higher positions in the reactivity series readily corrode upon contact with atmospheric air. The rate of corrosion decreases as we move down. What could be the reason? Examine in terms of reactivity.

Fig. 6.8

Even though aluminium is a metal at the top of the reactivity series, it resists corrosion to some extent. Why?

The oxide coatings on metals such as aluminium, zinc and tin are thin and nonporous and therefore protect the metal from further corrosion.

How does the corrosion of iron differ from this? The hydrated iron oxide (rust) that forms on the surface of iron is porous and powdery. So this process of corrosion continues until the metal is completely destroyed.

The factors given below influence the corrosion of metals.

- Nature of the metal
- Moisture in which gases or salts are dissolved
- Contact with other metals

How does the contact with other metals alter the rate of corrosion? Let us do an experiment (Figure 6.9).

Prepare a very dilute solution of sodium chloride in two large test tubes. Place iron nails, one tightly wrapped with copper wire and the other with magnesium wire. Observe the test tube for five to six days. Record the results of the observation in the science diary.

In which test tube, the iron nail undergoes rusting?

It is found that the iron nail in contact with copper corrodes easily while that in contact with magnesium is protected. Why does this happen?

When a metal comes in contact with another metal in the presence of an electrolyte, it acts as a simple voltaic cell.

When iron comes into contact with copper, which metal will release electrons? Why?

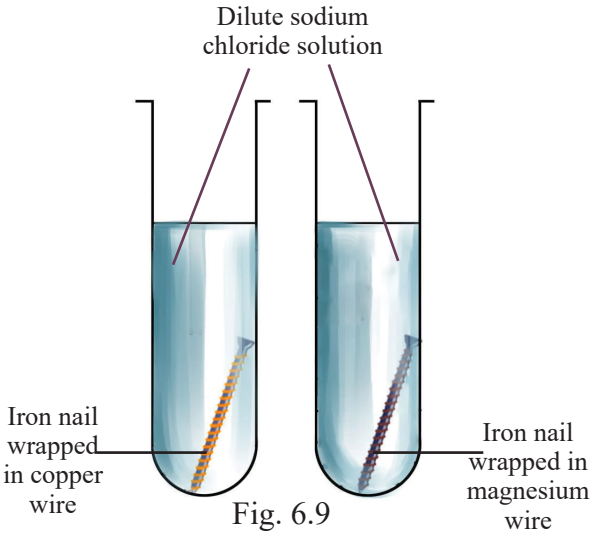


Fig. 6.9

The iron atoms, that lose electrons, change into ions. The rate of corrosion of iron increases. Here iron acts as anode and copper acts as cathode.

Why is iron protected when it is in contact with magnesium?

Isn't it because the more reactive magnesium donates electrons?

In the cell formed here, magnesium acts as anode and iron as cathode.

Which metal will corrode when two metals are in contact in the presence of moisture in the air?

The more reactive / The less reactive

It is seen that the metal acting as anode is destroyed and the metal acting as cathode is protected. This is an important method to prevent corrosion of iron. This is known as cathodic protection.

- Why is it not advisable to join aluminium wires and copper wires, and to join iron materials to copper piping for electrical purposes?
- Why are zinc or magnesium blocks always attached to sea bridges and ship hulls?

Discuss various methods to prevent corrosion of metals and list them in the science diary.

Metals play a crucial role in the human body also. We can learn more about it in the higher classes.

Bioinorganic chemistry

The presence of certain metals in humans, albeit in minute amounts, is essential for normal biological functions. Bioinorganic chemistry is a branch of chemistry that deals with the role of metals in biological processes. It can also be said that it is a branch of science that connects inorganic chemistry and biochemistry. This branch of chemistry deals with the movement and storage of metal ions in the human body, toxic properties of metal ions, metals with medicinal uses and much more.

Haemoglobin

Iron is one of the elements required for the day to day functioning of the human body. Haemoglobin is an iron containing protein molecule which carries oxygen. These pigments, found in the red blood cells in human body, carries oxygen from the lungs to cells and carbon dioxide from cells to the lungs.

Chlorophyll

Chlorophyll is a green pigment containing magnesium. This substance, found in the leaves of plants, is the basic factor in photosynthesis.

Let us assess

1. Define the following and answer the given questions.

   (1) Ore - What is the ore of aluminium?

   (2) Roasting - Which type of ores are subjected to roasting?

   (3) Reducing agent - What is the reducing agent used in the manufacture of aluminium?

   (4) Flux - Which flux is used in the manufacture of copper? Why?

   (5) Leaching - Which metal ore is leached with sodium cyanide?

2. Find the relation and write the answer.

   Zinc sulphide : Roasting Calcium carbonate : .....

   Magnetite : Magnetic separation Bauxite : .....

3. The chemical reaction that occurs when calcium carbonate is heated is given.

$$CaCO_3 \xrightarrow{heat} CaO + CO_2$$

   How is this chemical change of calcium carbonate utilised in the industrial preparation of iron?

4. Following are two facts related to the manufacture of an industrially important metal.

   - The ore is treated with hot NaOH solution.
   - Electricity is used as the reducing agent to extract the metal.

   (i) These facts are related to the production of which metal?

   (ii) What is the reason for using electricity as the reducing agent?

   (iii) Which substance is used as the electrolyte here?

   (iv) Which gas is liberated at the anode?

5. A portion of the periodic table is given.

	^{5}B	
	^{13}Al	^{14}Si
	^{31}Ga	

   (i) Which method given below is more possible for the production of gallium?

   (Reduction using carbon, electrolysis)

   (ii) If gallium chloride ($GaCl_3$) is electrolysed, what is the product formed at the cathode. Write the chemical equation.

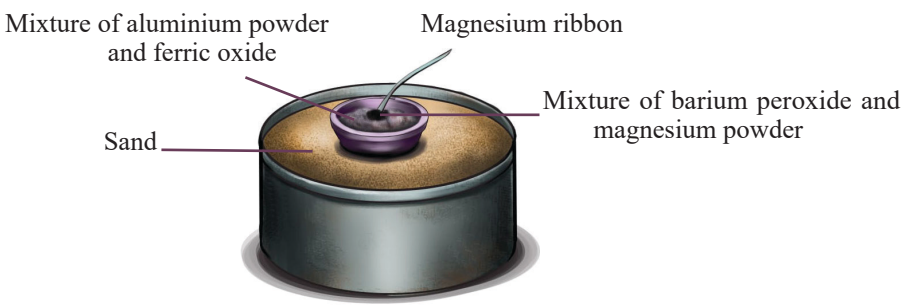
   (iii) Write the subshell electron configuration of the outermost shell of gallium.

6. Corrosion is a process by which iron is converted into its oxide. This is an oxidation reaction.
   (i) What is oxidation?
   (ii) Complete the following chemical equation.
   ..... Fe + ... O_2 → ..... Fe_2O_3
   (iii) Suggest two methods for preventing the corrosion of iron.
7. Alloys containing iron are given. Find a, b, c, d.

Alloys	Constituent elements	Uses
Alnico	(a)	(b)
(c)	Fe, Cr, Ni, C	Resist the corrosion of iron. Making of utensils
Silicon steel	Fe, Si, C	(d)

Extended activities

1. Thermite process



Mix equal amounts of anhydrous aluminium powder and iron oxide in a crucible. Immerse it in a tin pot filled with sand. Place a little of barium peroxide and magnesium powder mixture on top of the first mixture. Fix a 6 cm long polished magnesium ribbon in slanting position on its top. After lighting the ribbon, watch it from a distance.

After the reaction stops, examine the crucible and you will find a small globule of iron.

$$Fe_2O_3 + 2Al \xrightarrow{heat} Al_2O_3 + 2Fe$$

- Analyse the chemical changes that took place here based on the metallurgical process.
- Find instances where thermite process is used in daily life.
- Write the chemical equation for the production of chromium from chromic oxide (Cr_2O_3) in this way.

2. Prepare a note on how the rusted iron materials collected from the scrap iron shop are used in making steel. Present it in the class.
3. Prepare a flow chart showing the various steps in the purification of bauxite.